Vidyalankar Institute of Technology
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Department of Electronics and Telecommunication Engineering
(NBA ACCREDIATED)
Digital System Design Laboratory
Academic Year 2025-2026, SEM-5, Odd Semester

Course Code	EC24P
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Class	T.E – EXTC/EXCS SEM-V
Division	B
Date of Performance	1-8-25
Date of Submission	8-8-25

EXPERIMENT NO.3

Estimated time to complete this experiment: 02 hours.

Objective: To simulate and Implement Full Adder & Full Subtractor using all methods of modeling.

CO to be achieved: CO1, CO2

Expected Outcome of Experiment:

- Students will learn to design, simulate and synthesis the arithmetic circuit using VIVADO & VERILOG.

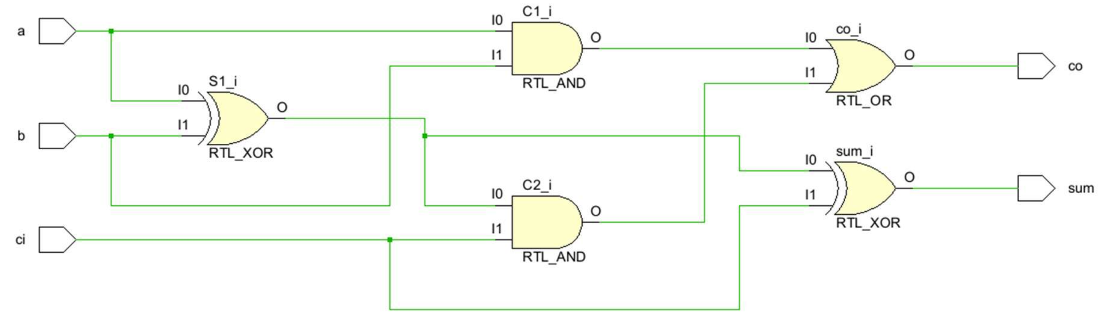
Pre Lab/ Prior Concept: Designing of Full adder and subtractor circuits.

Apparatus/TOOL:

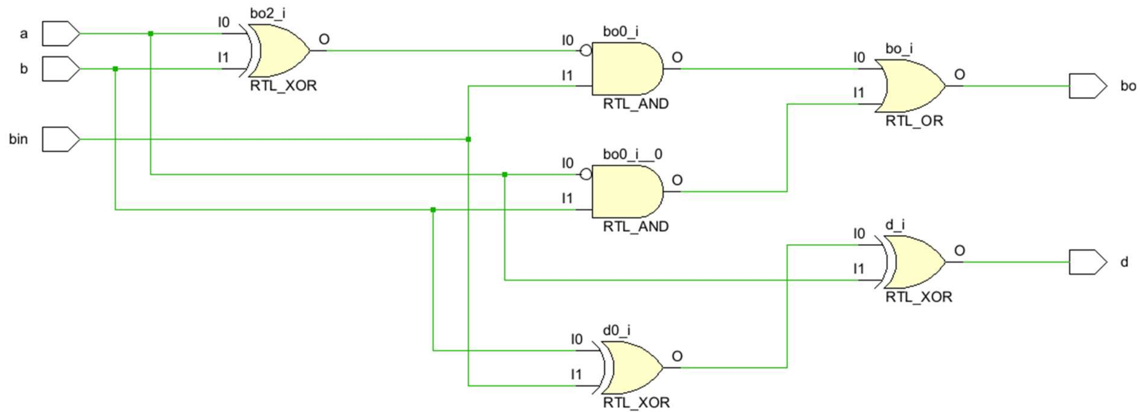
- PC
- VIVADO TOOL
- SPARTABN-7 FPGA Board.

CIRCUIT DIAGRAM.

Full Adder:



Full Subtractor:



VERILOG -CODE.

1. Full Adder

Gate-Level Modeling

Code

```
module Full_Adder_Gate_Level(output Sum, Carry, input A, B, Cin);
  wire S1, C1, C2;

  // Logic for Sum
  xor (S1, A, B);
  xor (Sum, S1, Cin);
```

```
  // Logic for Carry
  and (C1, A, B);
  and (C2, S1, Cin);
  or (Carry, C1, C2);
endmodule
```

Test Bench

```
module Full_Adder_Gate_Level_tb;
  reg A, B, Cin;
  wire Sum, Carry;

  Full_Adder_Gate_Level uut (.Sum(Sum), .Carry(Carry), .A(A), .B(B), .Cin(Cin));

  initial begin
    $monitor("A = %b, B = %b, Cin = %b, Sum = %b, Carry = %b", A, B, Cin, Sum, Carry);
    A = 0; B = 0; Cin = 0; #10;
    A = 0; B = 0; Cin = 1; #10;
    A = 0; B = 1; Cin = 0; #10;
    A = 0; B = 1; Cin = 1; #10;
    A = 1; B = 0; Cin = 0; #10;
    A = 1; B = 0; Cin = 1; #10;
    A = 1; B = 1; Cin = 0; #10;
    A = 1; B = 1; Cin = 1; #10;
    $stop;
  end
endmodule
```

Dataflow Modeling

Code

```
module Full_Adder_Dataflow(output Sum, Carry, input A, B, Cin);
  assign Sum = A ^ B ^ Cin;        // XOR operation for Sum
  assign Carry = (A & B) | (B & Cin) | (A & Cin); // Logic for Carry
endmodule
```

Behavioral Modeling

Code

```
module Full_Adder_Behavioral(output reg Sum, Carry, input A, B, Cin);
  always @(*) begin
    Sum = A ^ B ^ Cin;            // XOR operation for Sum
    Carry = (A & B) | (B & Cin) | (A & Cin); // Logic for Carry
  end
endmodule
```

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2. Full Subtractor

Gate-Level Modeling

Code

```
module Full_Subtractor_Gate_Level(output Diff, Borrow, input A, B, Bin);
  wire D1, B1, B2;

  // Logic for Difference
  xor (D1, A, B);
  xor (Diff, D1, Bin);

  // Logic for Borrow
  and (B1, ~A, B);
  and (B2, ~D1, Bin);
  or (Borrow, B1, B2);
endmodule
```

Test Bench

```
module Full_Subtractor_Gate_Level_tb;
  reg A, B, Bin;
  wire Diff, Borrow;

  Full_Subtractor_Gate_Level uut (.Diff(Diff), .Borrow(Borrow), .A(A), .B(B), .Bin(Bin));

  initial begin
    $monitor("A = %b, B = %b, Bin = %b, Diff = %b, Borrow = %b", A, B, Bin, Diff, Borrow);
    A = 0; B = 0; Bin = 0; #10;
    A = 0; B = 0; Bin = 1; #10;
    A = 0; B = 1; Bin = 0; #10;
    A = 0; B = 1; Bin = 1; #10;
    A = 1; B = 0; Bin = 0; #10;
    A = 1; B = 0; Bin = 1; #10;
    A = 1; B = 1; Bin = 0; #10;
    A = 1; B = 1; Bin = 1; #10;
    $stop;
  end
endmodule
```

Dataflow Modeling

Code

```
module Full_Subtractor_Dataflow(output Diff, Borrow, input A, B, Bin);
  assign Diff = A ^ B ^ Bin;              // XOR for Difference
  assign Borrow = (~A & B) | (~(A ^ B) & Bin); // Logic for Borrow
endmodule
```

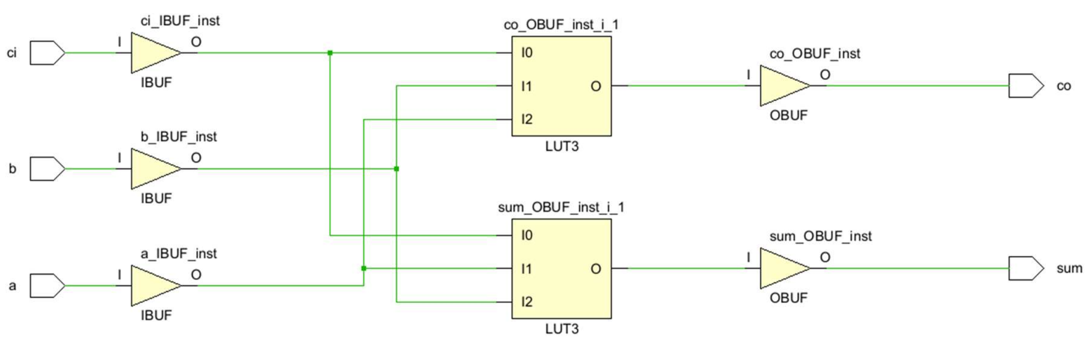
Behavioral Modeling

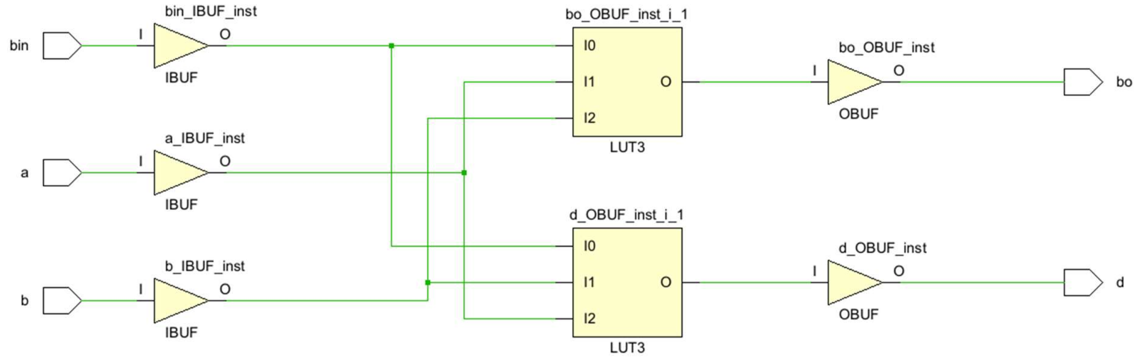
Code

```
module Full_Subtractor_Behavioral(output reg Diff, Borrow, input A, B, Bin);
  always @(*) begin
    Diff = A ^ B ^ Bin;                    // XOR for Difference
    Borrow = (~A & B) | (~(A ^ B) & Bin);   // Logic for Borrow
  end
endmodule
```

SIMULATION RESULTS.

SYNTHSIS MODEL.





CONCLUSION.

In this experiment, we successfully designed and simulated Full Adder and Full Subtractor circuits using all three modeling techniques in Verilog: Gate-Level, Dataflow, and Behavioral. Each method gave the correct output as expected. We also verified the functionality using testbenches and observed the simulation results. Through this practical, we learned how different Verilog modeling styles work and how to implement basic arithmetic circuits on the FPGA using the Vivado tool**.**

POST LAB QUESTION.

1.Derive the Boolean equation of sum and carry for full adder.

Let the inputs be:
A, B, Cin (carry-in)
Let the outputs be:
Sum, Carry

- Sum = A ⊕ B ⊕ Cin
  (⊕ is XOR operation)
- Carry = (A · B) + (B · Cin) + (A · Cin)
  (· is AND, + is OR)

2.Derive the Boolean equation of borrow and difference for full subtractor.

Let the inputs be:
A, B, Bin (borrow-in)
Let the outputs be:
Diff, Borrow

- Diff = A ⊕ B ⊕ Bin
- Borrow = (A.B)+(A ⊕ B) · Bin)

ASSESMENT RECORD.

Practical Performance [Model/Code, Results] **(5 Marks)**	**Presentation/ Learning Outcome** [Observation, Conclusion] **(5 Marks)**	**Total (10 Marks)**	**Sign**